Agile Process Development and DevOps Lab

Model Exam

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Problem Statement: 67

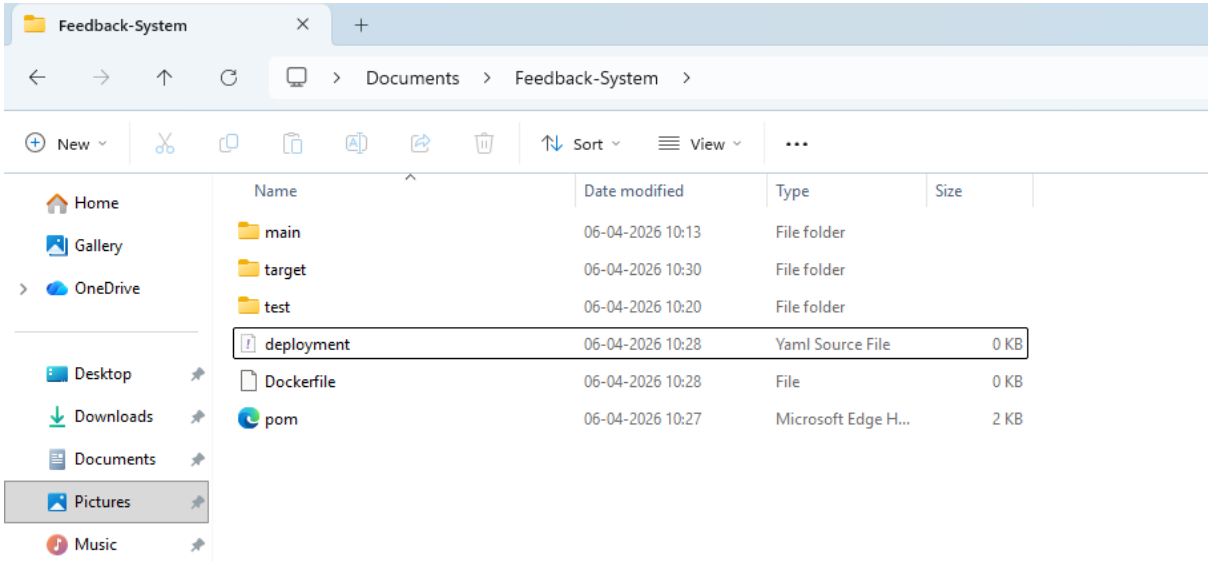
Feedback Collection System

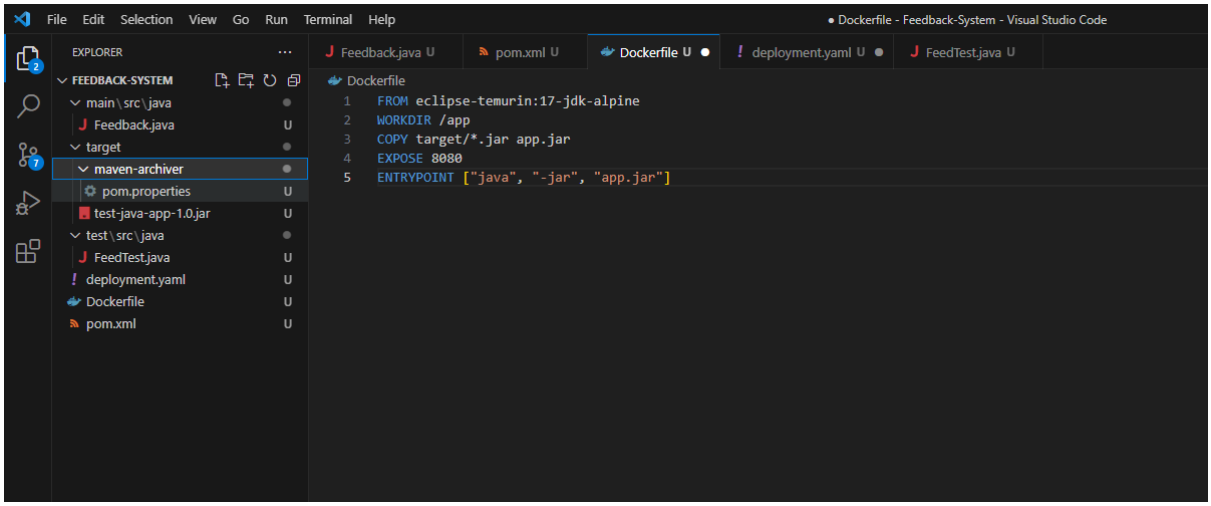
Ensure user feedback is recorded and stored correctly.

Tasks:

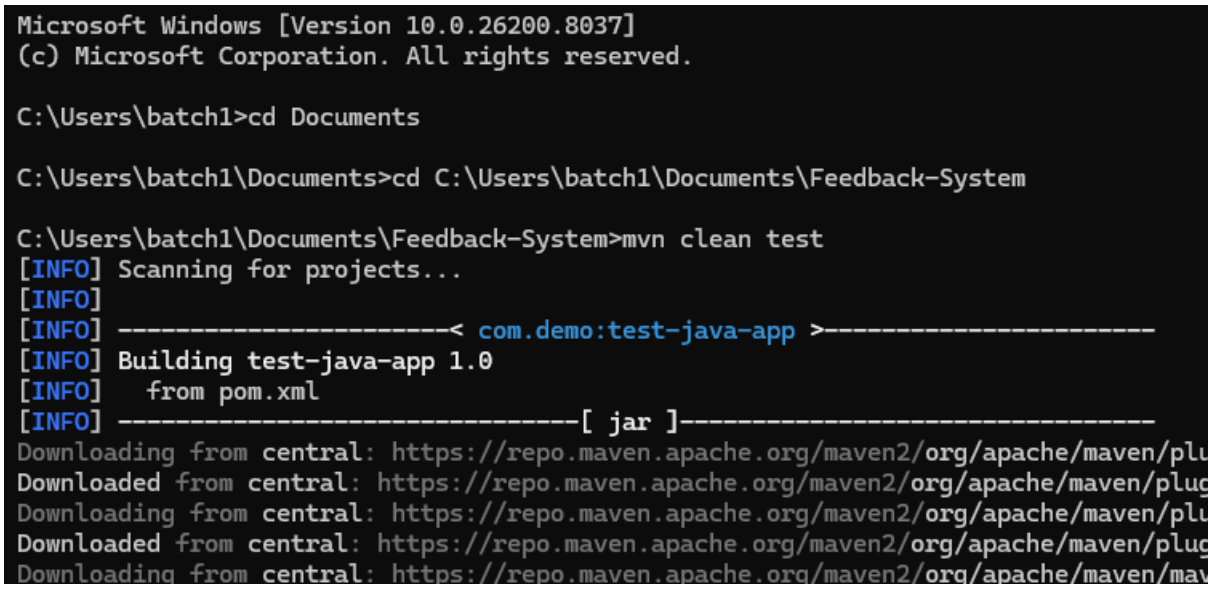
- Develop feedback submission logic
- Write JUnit tests for input validation
- CI/CD pipeline with automated testing
- Deploy microservice on Docker and Kubernetes

Folder structure





mvn clean test



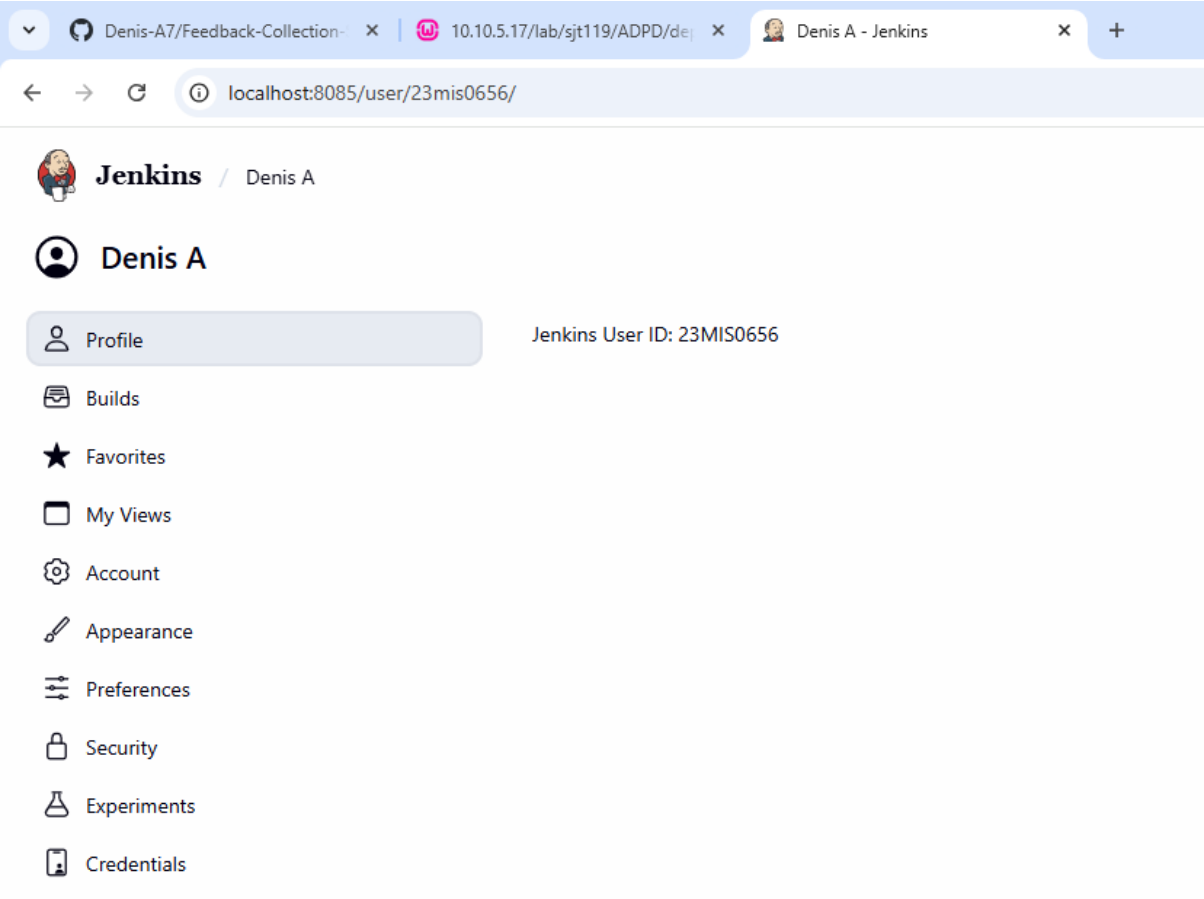
```
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/p
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/share
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.6/a
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/suref
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/sure
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aet
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/aether/aet
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/pl
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.6/as
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/suref
[INFO] No tests to run.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.104 s
[INFO] Finished at: 2026-04-06T10:29:59+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\batch1\Documents\Feedback-System>
```

mvn clean package

```
C:\Users\batch1\Documents\Feedback-System>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-j
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-p
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/4
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/42
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/32/apache-32.
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/32/apache-32.p
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.10.2/juni
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.10.2/junit
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-j
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ test-java-app ---
```

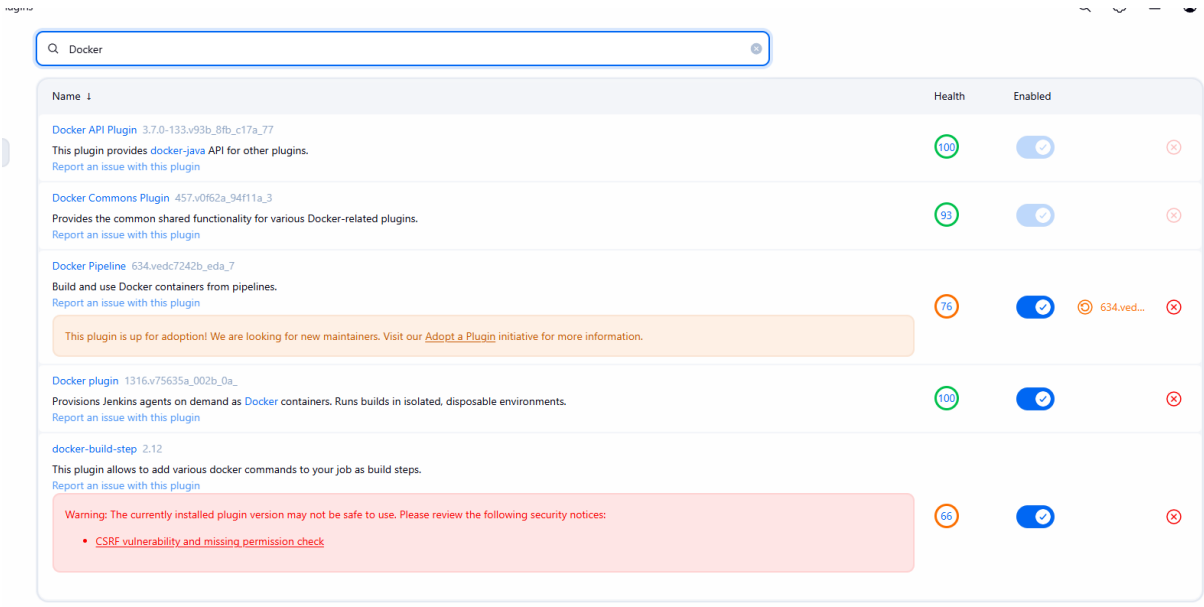
```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-
Downloaded from central: https://repo.maven.apache.org/maven2/commons-codec/commons-codec/
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.
[WARNING] JAR will be empty - no content was marked for inclusion!
[INFO] Building jar: C:\Users\batch1\Documents\Feedback-System\target\test-java-app-1.0.ja
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.385 s
[INFO] Finished at: 2026-04-06T10:30:48+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\batch1\Documents\Feedback-System>
```

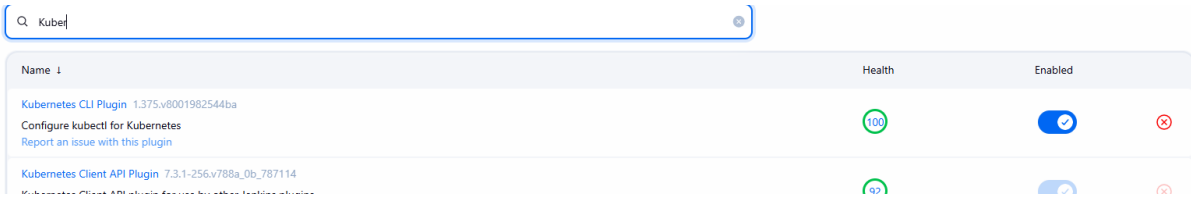
In my Jenkins



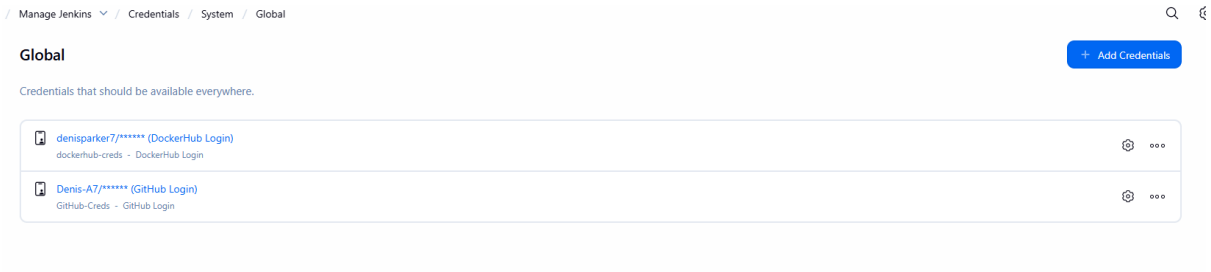
Check Installed plugins:

1. Docker Plugin
2. Docker Pipeline
3. Kubernetes CLI

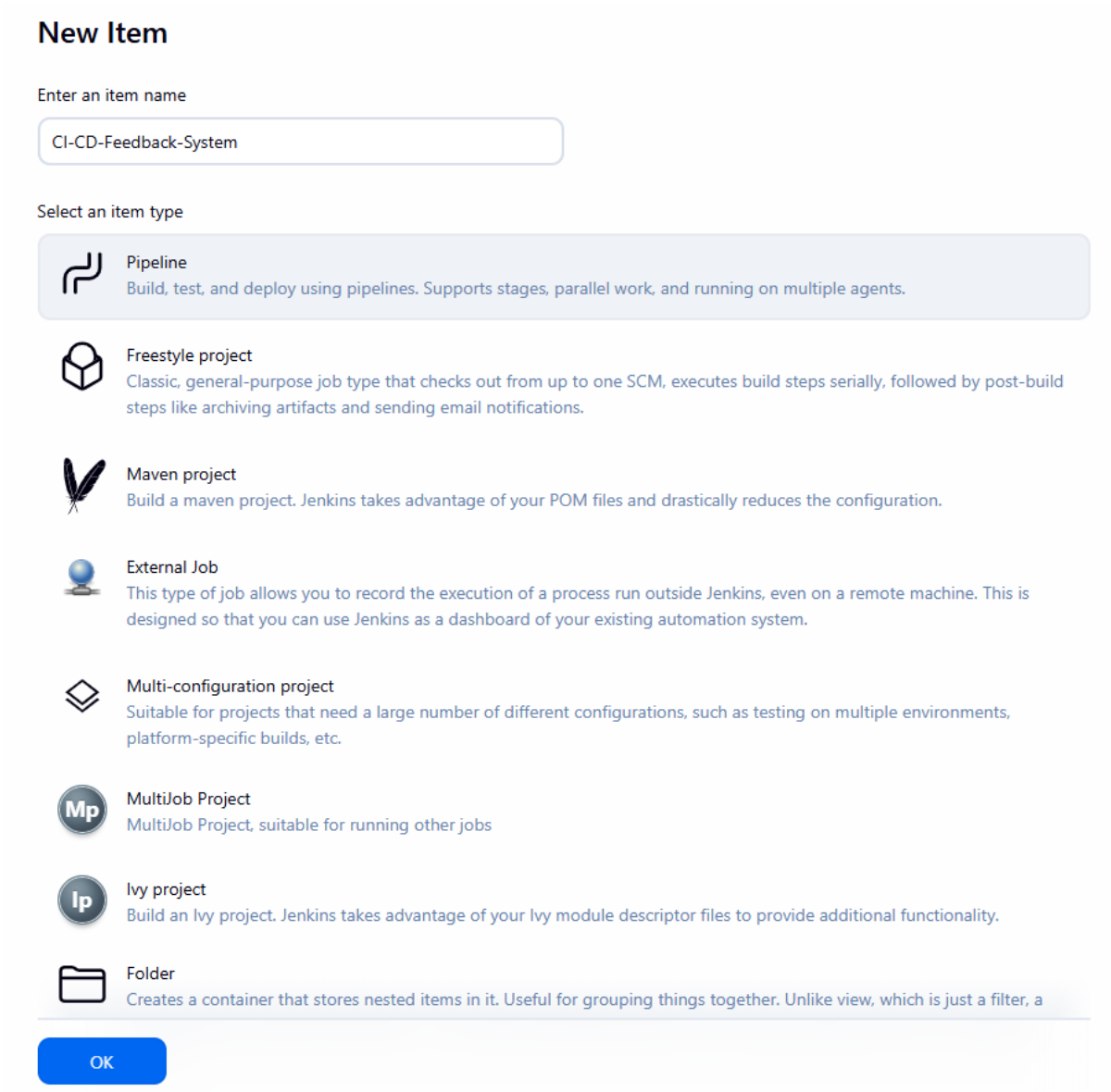




Set Credentials in Jenkins to be used by the Jenkins pipeline



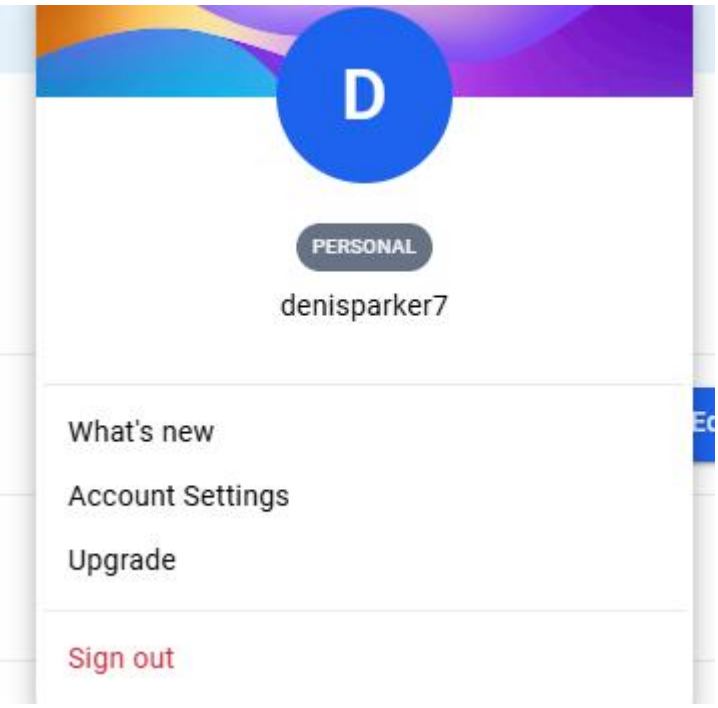
Create a new Pipeline

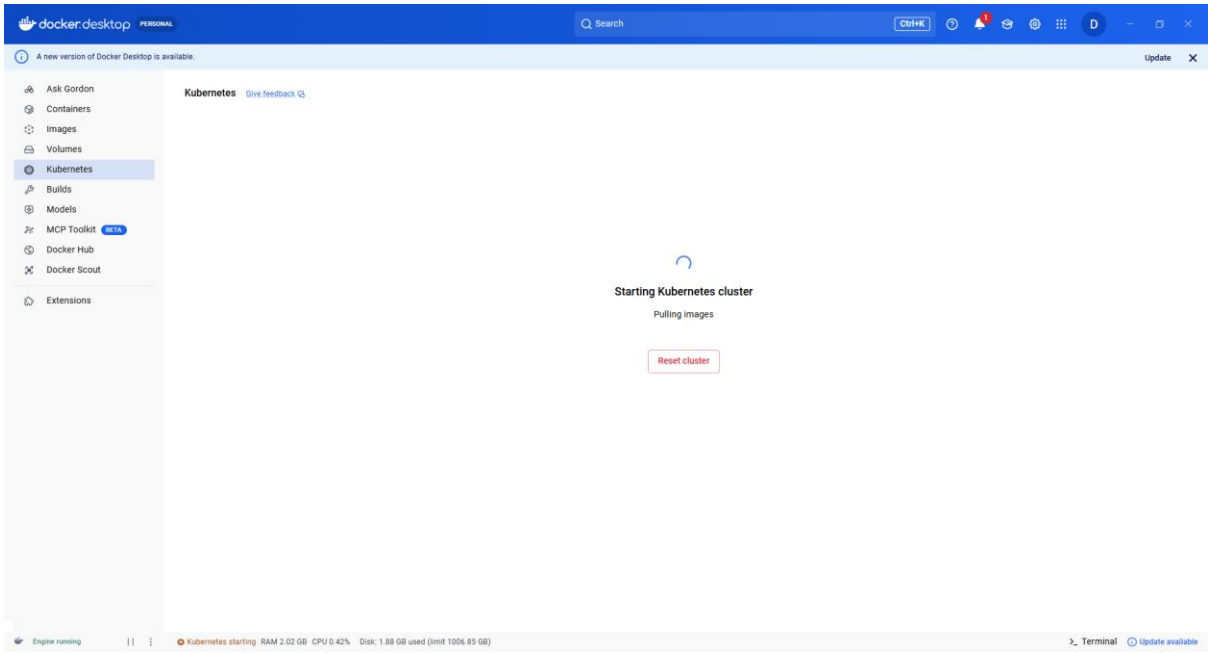


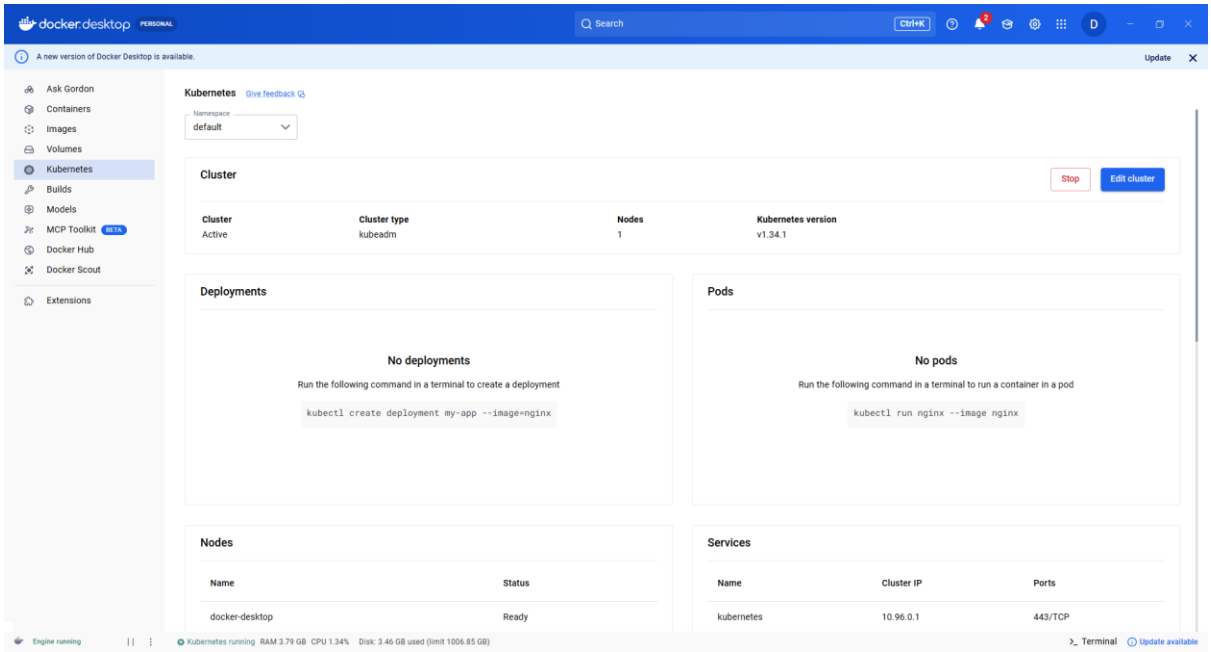
In Docker Desktop

Go to Kubernetes and create a cluster with 1 Node

My username in Docker Desktop







Cluster created

Currently running services

Services

Name	Cluster IP	Ports
kubernetes	10.96.0.1	443/TCP

Check it in Command Prompt

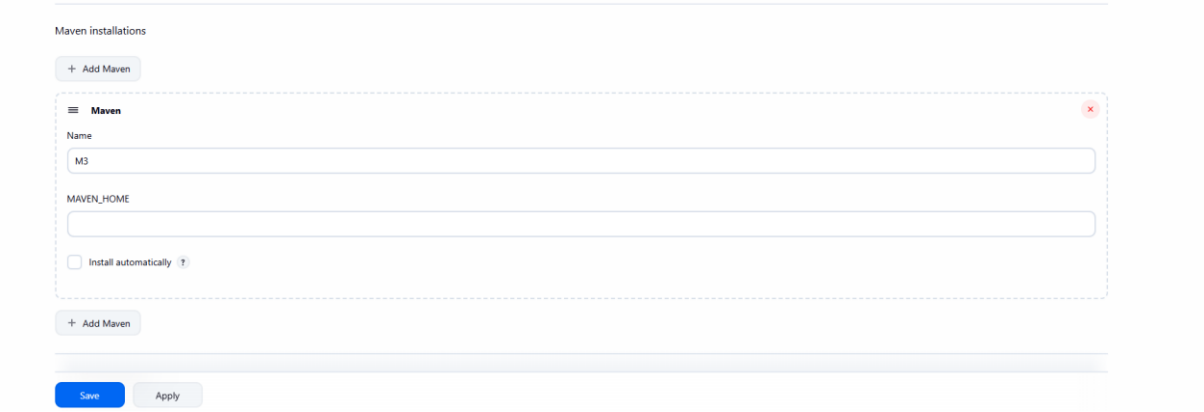
-kubectl get svc

```
C:\Users\batch1\Documents\Feedback-System>kubectl get svc
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes    ClusterIP   10.96.0.1    <none>        443/TCP   8m2s

C:\Users\batch1\Documents\Feedback-System>
```

Check if maven is installed

```
C:\Users\batch1\Documents\Feedback-System>mvn -v
Apache Maven 3.9.12 (848fbb4bf2d427b72bdb2471c22fced7ebd9a7a1)
Maven home: C:\Program Files\Java\apache-maven-3.9.12
Java version: 21.0.9, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"

C:\Users\batch1\Documents\Feedback-System>
```

Maven is not configured in Jenkins



So the pipeline script has failed

Pipeline Script:

```
pipeline {

   agent any


   tools {

      // Install the Maven version configured as "M3" and add it to the path.

      maven "M3"

   }


   stages {

      stage('Build') {

         steps {
```

```
            // Get some code from a GitHub repository
            git 'https://github.com/Denis-A7/Feedback-Collection-System.git'

            // Run Maven on a Unix agent.
            sh "mvn -Dmaven.test.failure.ignore=true clean package"

            // To run Maven on a Windows agent, use
            // bat "mvn -Dmaven.test.failure.ignore=true clean package"
        }

        post {
            // If Maven was able to run the tests, even if some of the test
            // failed, record the test results and archive the jar file.
            success {
                junit '**/target/surefire-reports/TEST-*.xml'
                archiveArtifacts 'target/*.jar'
            }
        }
    }
  }
}
```

Script ?

```
pipeline {
    agent any

    tools {
        // Install the Maven version configured as "M3" and add it to the path.
        maven "M3"
    }

    stages {
        stage('Build') {
            steps {
                // Get some code from a GitHub repository
                git 'https://github.com/Denis-A7/Feedback-Collection-System.git'

                // Run Maven on a Unix agent.
```

Use Groovy Sandbox ?

Pipeline Syntax

#1

#1 (Apr 6, 2026, 11:05:44 AM)

Started by user Denis A

This run spent:

- 19 ms waiting;
- 0.48 sec build duration;
- 0.5 sec total from scheduled to completion.

No changes.

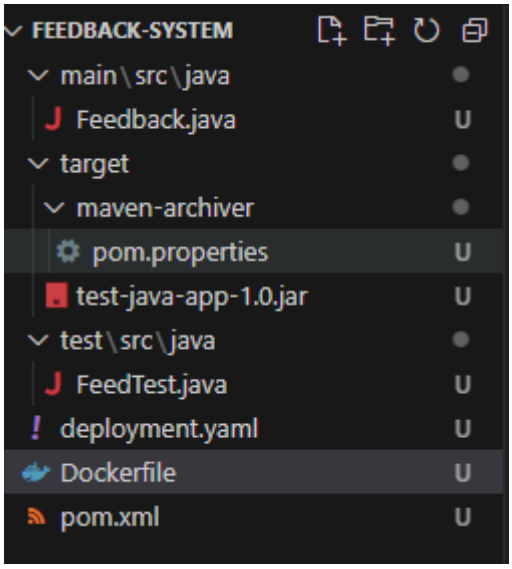
Console Output

Download | Copy | View as plain text

```
Started by user Denis A
org.codehaus.groovy.control.MultipleCompilationErrorsException: startup failed:
WorkflowScript: 6: Tool type "maven" does not have an install of "M3" configured - did you mean "null"? @ line 6, column 15.
           maven "M3"
                 ^

1 error

        at org.codehaus.groovy.control.ErrorCollector.failIfErrors(ErrorCollector.java:309)
        at org.codehaus.groovy.control.CompilationUnit.applyToPrimaryClassNodes(CompilationUnit.java:1107)
        at org.codehaus.groovy.control.CompilationUnit.doPhaseOperation(CompilationUnit.java:624)
        at org.codehaus.groovy.control.CompilationUnit.processPhaseOperations(CompilationUnit.java:602)
        at org.codehaus.groovy.control.CompilationUnit.compile(CompilationUnit.java:579)
        at groovy.lang.GroovyClassLoader.doParseClass(GroovyClassLoader.java:323)
        at groovy.lang.GroovyClassLoader.parseClass(GroovyClassLoader.java:293)
        at PluginClassLoader for script-security//org.jenkinsci.plugins.scriptsecurity.sandbox.groovy.GroovySandbox$Scope.parse(GroovySandbox.java:162)
        at PluginClassLoader for workflow-cps//org.jenkinsci.plugins.workflow.cps.CpsGroovyShell.doParse(CpsGroovyShell.java:202)
        at PluginClassLoader for workflow-cps//org.jenkinsci.plugins.workflow.cps.CpsGroovyShell.reparse(CpsGroovyShell.java:186)
        at PluginClassLoader for workflow-cps//org.jenkinsci.plugins.workflow.cps.CpsFlowExecution.parseScript(CpsFlowExecution.java:669)
        at PluginClassLoader for workflow-cps//org.jenkinsci.plugins.workflow.cps.CpsFlowExecution.start(CpsFlowExecution.java:615)
        at PluginClassLoader for workflow-job//org.jenkinsci.plugins.workflow.job.WorkflowRun.run(WorkflowRun.java:341)
        at hudson.model.ResourceController.execute(ResourceController.java:101)
        at hudson.model.Executor.run(Executor.java:460)
Finished: FAILURE
```

Since Maven is not configured Jenkins Pipeline cannot run the build successfully

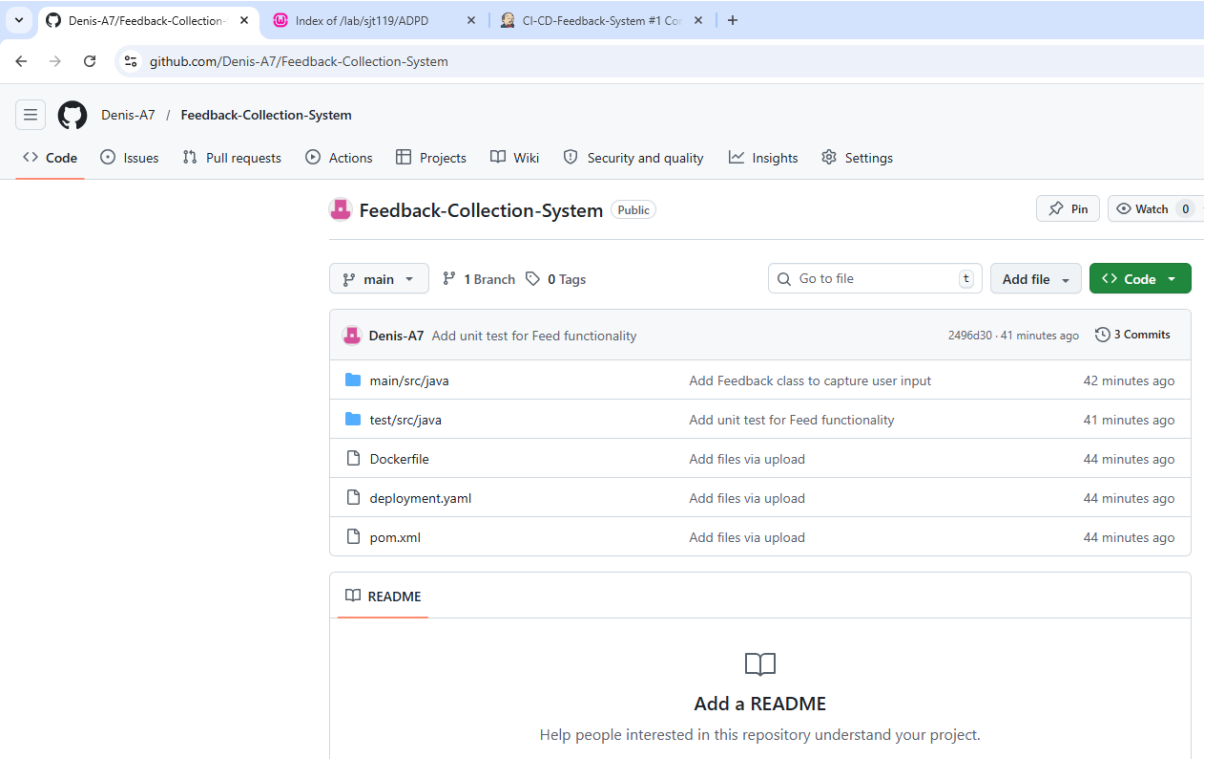
So the local maven test is used above already.



The .jar file created after running mvn test package

If the pipeline worked, then the Docker image will be created using our Docker Credentials and the image will be pushed to DockerHub and the microservice will be deployed using Kubernetes Cluster.

Git Hub Repository



https://github.com/Denis-A7/Feedback-Collection-System.git

Thank You!